

EXPERIMENT 27: DO CROPPING, COPYING AND PASTING IMAGE INSIDE ANOTHER IMAGE USING OPENCV

Aim

To crop a selected region from an image, copy it, and paste it inside another image using Python and OpenCV.

Algorithm

1. Import the OpenCV library.
2. Read the source and destination images.
3. Select the coordinates of the region to be cropped.
4. Crop the selected region from the source image.
5. Select the position in the destination image where the cropped region has to be pasted.
6. Copy the cropped region into the selected destination area.
7. Save and display the resulting image.

Python Code

```
import cv2

# Read source image
source = cv2.imread("source.jpg")

# Read destination image
destination = cv2.imread("destination.jpg")

if source is None or destination is None:
    print("Image not found")
    exit()

# Get image dimensions
height, width = source.shape[:2]

# Define crop coordinates
```

```
x1 = int(width * 0.25)
y1 = int(height * 0.20)
x2 = int(width * 0.65)
y2 = int(height * 0.70)

# Crop the selected region
cropped = source[y1:y2, x1:x2]

# Define paste position
paste_x = 20
paste_y = 20

# Get cropped image dimensions
crop_height, crop_width = cropped.shape[:2]

# Copy and paste the cropped region
destination[
    paste_y:paste_y + crop_height,
    paste_x:paste_x + crop_width
] = cropped

# Save output
cv2.imwrite("crop_copy_paste_output.jpg", destination)

# Display images
cv2.imshow("Source Image", source)
cv2.imshow("Cropped Image", cropped)
cv2.imshow("Final Image", destination)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input – Source Image



Cropped Region



Input – Destination Image



Output – Cropped Region Pasted Inside Another Image



Result

Thus, the selected region was successfully cropped from the source image, copied, and pasted inside another image using OpenCV.